Build a complete prediction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use one diagram or set of equations to connect net force, change in velocity, and final position.

SUCCESS CRITERIA - I can...

1. Connect force, acceleration and position consistently.
2. Show quantities and units at every stage.
3. Explain what the prediction treats as true and when it would stop working.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “reference frame” and “net force” to explain your answer. Sentence starters are optional.

reference frame: the origin and directions used to describe motion

net force: the vector sum of all forces on one object

prediction: what a model says may happen next

01 Warm-up: bring back an earlier idea

Maximum static friction is **50 N**. A **20 N** push leaves a box still. **Find actual friction**. A **60 N** push starts sliding with **40 N** kinetic friction; **find net force**.

Word help: friction = a contact force opposing relative sliding or its tendency

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

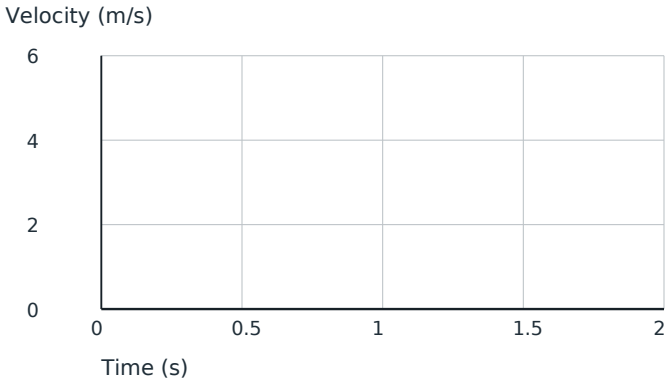
02 Worked example: explain the prediction

Reference: $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$; $x_f = x_i + \frac{1}{2}(v_i + v_f)\Delta t$. Constant acceleration; use one frame. i = initial; f = final. Apply each direction separately.

Worked reasoning: A **2 kg** cart starts from rest under **6 N** right and **2 N** left for **2 s**. Net force **4 N** gives $a = \mathbf{2\ m/s^2}$. Its displacement is $\frac{1}{2} \times \mathbf{2} \times \mathbf{2^2} = \mathbf{4\ m}$ right.

Your turn: 1. **Draw the force diagram** and the matching velocity-time triangle.
2. **Check the 4 m** using its area.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

Build a complete prediction

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Start with forces

A **2 kg** cart starts at rest. There are **6 N** right and **2 N** left. **Find net force** and acceleration.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.



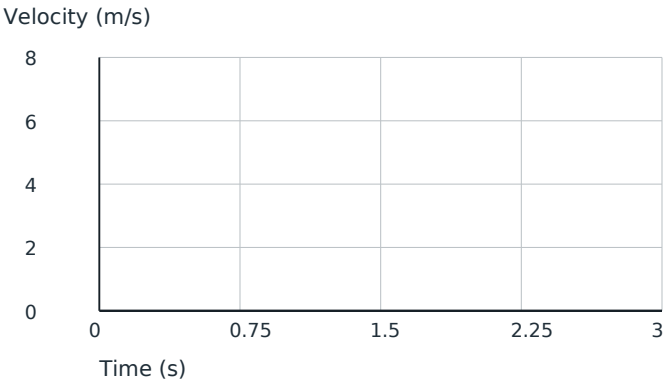
The dot represents the object. Mass 2 kg; starts at rest

Marzano check - circle: 1 2 3

04 Predict motion

The forces on the cart in question **3** stay constant for **3 s**.
1. **Draw its velocity-time graph**, then **find final velocity**, average velocity and displacement.
2. If the cart starts at the **1 m** mark, where does it finish?

Sentence starter: I found ... [unit]. This value means ... in this situation.

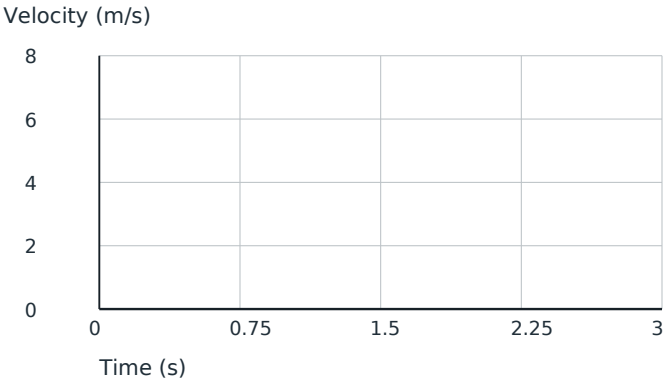


Marzano check - circle: 1 2 3

05 Explain a limit

Your prediction in question **4** treats net force as constant.
1. **Name a change** that could make the net force vary.
2. **Draw how the velocity**–time line would change if a larger, constant friction force opposed the motion throughout.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3

Build a complete prediction

antocciphysics

Continue down each column.

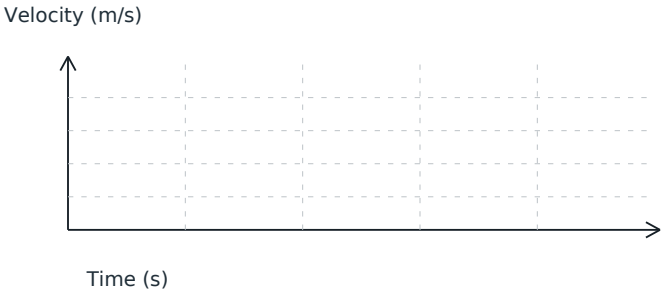
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A cart accelerates uniformly from rest to **14 m/s** *east* in **5 s**. **Find acceleration**, average velocity and displacement.
Sketch the velocity–time graph; **label the given values**.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

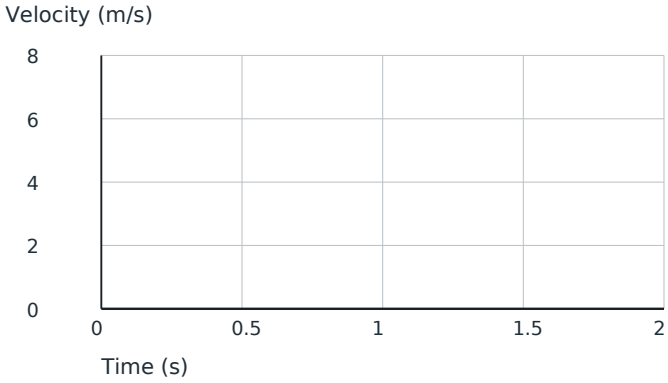


Marzano check - circle: 1 2 3

07 Exit check

A **3 kg** rover starts from rest with a constant **9 N** net force to the right for **2 s**.
1. **Find acceleration** and final velocity.
2. **Draw a graph** that checks your result, and **state one thing you** are treating as true.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**. A second cart must cover the same **4 m** in **2 s** from rest but has twice the mass. What net force is required? **Show your chain of reasoning**.

Sentence starter: I notice ... . This supports ... because ... .

Add and label force arrows.

The second cart has twice the worked example's mass.

Marzano check - circle: 1 2 3